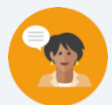


14.1 Constructing Two-Way Tables

Lesson Introduction

 Benchmarks	MA.912.DP.3.1 Construct a two-way frequency table summarizing bivariate, categorical data. Interpret joint and marginal frequencies and determine possible associations in terms of a real-world context.			 Learning Targets	- I can construct a two-way frequency table. - I can determine joint and marginal frequencies in a two-way frequency table.
	MA.912.DP.1.1 Given a set of data, select an appropriate method to represent the data, depending on whether it is numerical or categorical data, and on whether it is univariate or bivariate.				
	MA.912.DP.1.2 Interpret data distributions represented in various ways. State whether the data is numerical or categorical, whether it is univariate or bivariate, and interpret the different components and quantities in the display.				
 Benchmark Coherence	Building On		Working Toward		
	N/A		N/A		
 Essential Questions	- How can categorical bivariate data be organized? - How does a two-way frequency table organize data?				
 Lesson Narrative	In this lesson, students are introduced to two-way frequency tables and how to interpret the information they display. Students are expected to know the key terms and be able to differentiate between marginal and joint frequencies.				
 Component Summary	14.1.1 Warm-Up (~5 min.)	Review of percent calculations to prepare for this unit (<i>SE p. 305</i>)	14.1.4 Guided Practice (5 min.)	Students complete a two-way frequency table and interpret two data points (<i>SE p. 308</i>)	
	14.1.2 Guided Instruction (10 min.)	An overview of two-way frequency tables, defining key terms (<i>SE p. 306</i>)	14.1.5 Try It (10 min.)	A two-way frequency table is constructed and interpreted (<i>SE p. 309</i>)	
	14.1.3 Collaboration (10 min.)	Partners work to correct false statements (<i>SE p. 308</i>)	14.1.6 Wrap-Up (~5 min.)	Students reflect on their understanding (<i>SE p. 310</i>)	
 Resources	Glossary		Materials		Additional Preparation
	<u>Key Terms:</u> - Two-Way Frequency Table - Marginal Frequency - Joint Frequency - Joint Relative Frequency <u>Additional Terms:</u> N/A		N/A		Consider creating small intervention groups for students who are not proficient in the percentage questions from 14.1.1 Warm-Up, as these are prerequisite skills for Unit 14.

 Teacher Tips	Common Misconceptions	Things to Consider
	- Students may confuse joint and marginal frequencies. - Students may mistakenly identify the lower right box in the two-way frequency table as a marginal frequency. This number represents the total number of people surveyed and is not a marginal frequency.	Two-way frequency table and joint frequency table can be used interchangeably.
	Literacy and Language Routines	Mathematical Thinking and Reasoning (MTR) Standard
	Clarify, Critique, Correct Question 1 in 14.1.3 Collaboration allows students to engage in a collaboration on critiquing and correcting false statements. Students work together to decide if each statement about the two-way frequency table is true or false. Then they correct the false statements to make them true.	MA.K12.MTR.7.1: Real-World Contexts Clarifications in this standard explain that, “mathematicians who apply mathematics to real-world contexts use models and methods to understand, represent, and solve problems.” Throughout the lesson students learn what a two-way frequency table is and how to create and interpret one to represent real-world data.
	Consider previewing the key terms and additional terms, listed above, with struggling students, students with special needs, and/or ELL students. It may be helpful to shorten the lesson for students who need more time to complete their work than what is typically needed. One place where there is an opportunity to eliminate some questions without sacrificing content is 14.1.2 Guided Instruction, after the Teen Social Media Survey two-way frequency table. Questions 3-5 all ask students to circle numbers in the table, the marginal frequencies. Some of these questions could be eliminated. However, it is important for students to know where all of the marginal frequencies in the table are located. Another place in the lesson where there is an opportunity to shorten the material for students who need more time than what is typical is in 14.1.5 Try It. In this component, students are asked to construct two joint frequency tables, and are asked follow-up questions about the first table they construct. Eliminating the second question where students construct a two-way frequency table can help those students who need more time than what is typically needed get their work done in a reasonable amount of time. For students who need more of a challenge, consider asking them to rewrite any of the two-way frequency tables using joint relative frequencies.	
Lesson Notes:		

14.1.1 Warm-Up: Percent Calculations

Component Overview: Students solve problems involving percent. Provide students the time to complete the questions and the opportunity to check their work for correctness. Consider forming small intervention groups for students who are not proficient at this prerequisite skill.



14.1.1 Warm-Up: Percent Calculations

1. Newborn babies typically lose some weight in the first week after birth. Weight loss of up to 10% is common. If a newborn baby weighs 8.5 pounds at birth and loses 8% of their weight in the first week of life, how many pounds does the baby lose?

$$8.5 \cdot 0.08 = 0.68$$

A baby who weighs 8.5 pounds at birth and loses 8% of their body weight in the first week of life, loses 0.68 pounds.

2. Anya earned a score of 75% on her last math quiz. If she earned 18 points, how many total points was the quiz worth?

$$\frac{75}{100} = \frac{3}{4} = \frac{18}{24}$$

If Anya earned 18 points on her last math quiz and this is a score of 75%, the quiz was worth 24 points.

3. Five hundred gallons of maple sap were boiled to become 12.5 gallons of maple syrup. What percentage of the 500 gallons of sap is the 12.5 gallons of maple syrup?

$$\frac{12.5}{500} = 12.5 \div 500 = 0.025$$

The 12.5 gallons of syrup is 2.5% of the 500 gallons of sap.



Another possible way to structure 14.1.1 Warm-Up is to have students answer the questions independently, then check their work with their partner. If there are any questions they disagree with their partner on, those would be good to discuss as a class.

14.1.2 Guided Instruction: Two-Way Frequency Tables

Component Overview: This student-led 14.1.2 Guided Instruction provides students the opportunity to determine what type of data can be displayed in a two-way frequency table. New terms are introduced, and students are provided with multiple examples of each.



14.1.2 Guided Instruction: Two-Way Frequency Tables

A 2018 Pew Research Center survey of American teenagers was conducted to analyze trends in social media and technology.

741 teens who use social media were asked what effect they believe social media has had on people their age (mostly positive, mostly negative, neutral).

1. The effect of social media is a
 - ☐ quantitative variable.
 - ☒ categorical variable.

The data when asked what effect they believe social media has had on people their age (mostly positive, mostly negative, neutral) is summarized.

- 219 said the effect was mostly positive
- 211 said the effect was mostly negative
- 311 said the effect was neutral

On the same survey, the Pew Research Center also collected data from teens on how much time they believe they spend with their friends in person (too much, too little, or about the right amount).

2. Time spent with friends in person (too much, too little, or just the right amount) is a
 - ☐ quantitative variable.
 - ☒ categorical variable.



Students may need a reminder about the difference between quantitative and categorical data.

A **two-way frequency table** is a display used to summarize bivariate categorical data. The table can also be called a contingency table or a joint frequency table.

14.1.2 Guided Instruction: Two-Way Frequency Tables (continued)

The two-way frequency table displays the results for teens when asked about the effect of social media, in addition to the results when being asked about time spent with friends in person.

Teen Social Media Survey

		Effect on People Your Age		
		Mostly Positive	Mostly Negative	Neutral
Time spent With friends in Person	Too Much	22	21	20
	Too Little	87	77	105
	Just Right	110	113	186
	Total	219	211	311
		741		

- Circle the frequency for those who said the effect of social media was mostly positive.
There were 219 people who said the effect of social media was mostly positive.
- Circle the frequency for those who said the effect of social media was mostly negative.
The number of people who said the effect of social media was mostly negative was 211.
- Circle the frequency for those who said the effect of social media was neutral.
311 people said the effect of social media was neutral.
- How many teens reported spending too much time with friends in person? Circle the value in the table.
The number of teens who reported spending too much time with friends in person is 63.



In addition to teaching the mathematical definitions of joint and marginal frequency, consider connecting these names to their location in the table:

- Marginal frequencies are located in the margins of the table.
- Joint frequencies are located in places where two variables meet or are “joined”. They are found at the “joint” of two variables.



Consider reducing the content of this component for students who need longer than what is typical. Possible ways to shorten this component are described in the Differentiate Instruction section on page 1349.



Consider discussing the numbers in the table to help students solidify their understanding of how the numbers within a row and column are related. Some possible discussion questions follow:

- What do you notice about the numbers in the top row of the table?
- Where are some other places in the table where the numbers are related in this way?
- If only the marginal frequency of the column is missing, how can you find it?
- If only one joint frequency is missing from a row, how can you use the other three numbers to find the missing joint frequency?
- How can the joint frequencies be used to find the total?

14.1.2 Guided Instruction: Two-Way Frequency Tables (continued)

7. How many teens reported spending too little time with friends in person? Circle the value in the table.

The number of teens who reported spending too little time with friends in person is 269.

8. How many teens reported spending just the right amount of time with friends in person? Circle the value in the table.

409 teens reported spending just the right amount of time with friends in person.

Each of the values circled is a **marginal frequency**.

The total frequency for one category of a categorical variable, which is the total frequency for any row or column in a two-way frequency table, is called a **marginal frequency**.

9. How many people reported that the effect of social media is mostly positive **and** reported that they spend too much time with friends in person?

The number of people who reported that the effect of social media is mostly negative and they spend too much time with friends in person is 22.

10. How many people reported that the effect of social media is mostly negative **and** reported that they spend too much time with friends in person?

21 people reported that the effect of social media is mostly negative and they spend too much time with friends in person.

11. How many people reported that the effect of social media is mostly negative **and** reported that they spend too little time with friends in person?

The number of people who reported that the effect of social media is mostly negative and they spend too little time with friends in person is 77.



Students may believe that 741 is a marginal frequency. Since this is the total number of people surveyed, it is not considered a marginal frequency or a joint frequency.



Answering questions that can be looked up in the table is a good way to create more entry-points into the lesson.

14.1.2 Guided Instruction: Two-Way Frequency Tables (continued)



In a two-way table, a **joint frequency** is the number of times a combination of two conditions occurs.

12. Draw a box around each joint frequency in the table.
There is a red box around each joint frequency in the table.
13. What percentage of respondents reported that the effect of social media is mostly positive **and** reported that they spend too much time with friends in person?

$$\frac{\boxed{22}}{\boxed{741}} = 22 \div 741 = 0.030 = 3\%$$

Three percent of respondents reported that the effect of social media is mostly positive, and they spend too much time with friends in person.

14. What percentage of respondents reported the effect of social media is mostly negative **and** reported that they spend too much time with friends in person?

$$\frac{\boxed{21}}{\boxed{741}} = 21 \div 741 = 0.028 = 2.8\%$$

The percentage of respondents who reported that the effect of social media is mostly negative, and they spend too much time with their friends in person is 2.8%.



In a two-way table, a **joint relative frequency** is the ratio of the joint frequency and the total number of data points.



After students have answered questions 1-10 and read the definitions of marginal and joint frequencies, consider asking the class to discuss the similarities and differences between joint and marginal frequencies. Here are some possible key points to highlight:

- Marginal frequencies can be broken down into joint frequencies.
- Joint frequencies represent a more specific group of the data than marginal frequencies do.
- Joint frequencies break the data down into smaller groups than marginal frequencies do.



Joint relative frequency is formally introduced. This concept will be extended in the honors lessons. However, students may be confused about the difference between a joint frequency and a joint relative frequency. One way to explain this difference is by focusing on the word “**relative**”. A joint relative frequency is a joint frequency that is reported in comparison, or relation, to the rest of the data.



Discussion provides an auditory entry-point. Consider utilizing sentence stems, and provide ample time for students to grapple with academic language:

- _____ percent of respondents reported that the effect of social media is _____ and they spend _____ time with friends in person.
- The percentage of respondents who reported that the effect of social media is _____, and they spend _____ time with their friends in person is _____.

14.1.3 Collaboration: Interpreting a Two-Way Table

Component Overview: Students will work with their partner to decide if several statements about a two-way frequency table are true or false. They will also correct the false statements to make them true. This is a student-led component.



14.1.3 Collaboration: Interpreting a Two-Way Table

- With your partner, use the two-way frequency table to determine if each statement in the table is true or false. If a statement is false, correct the statement to make it true.

Teen Social Media Survey

Effect on People Your Age

Time spent With Friends in Person		Mostly Positive	Mostly Negative	Neutral	Total
	Too Much	22	21	20	63
	Too Little	87	77	105	269
	Just Right	110	113	186	409
	Total	219	211	311	741

269 participants reported that they spend too little time with friends in person.	☐ True ☐ False
More teens reported that they spend too little time with friends than just the right amount with friends in person. <i>More teens reported that they spend just the right amount of time with friends in person than too little time with friends in person.</i>	☐ True ☐ False
211 participants reported the effect of social media was mostly negative and spend just the right amount of time with friends in person. <i>113 participants reported the effect of social media was mostly negative and spend just the right amount of time with friends in person.</i>	☐ True ☐ False
186 participants reported that social media has a neutral effect and they spend just the right amount of time with their friends.	☐ True ☐ False
There was a total of 1,823 participants in the survey. <i>There was a total of 741 participants in the survey.</i>	☐ True ☐ False



Consider giving students the opportunity to share how they corrected the false statements. There are several different ways to correct the false statements, and students could benefit from hearing how other people corrected the false statements. Students could either meet with another pair of students or share their corrections with the class.



What is the most efficient way to correct the false statements?



For students who finish early, challenge them to determine how the false statements may have been developed. Have them determine how that person may have been thinking to make the false statement.

14.1.4: Guided Practice: Constructing a Two-Way Table

Component Overview: In this student-led 14.1.4 Guided Practice component, students work with their partner to construct a joint frequency table and classify data points as joint or marginal frequencies.



14.1.4 Guided Practice: Constructing a Two-Way Table

1. Kieran decides to gather data on hair and eye color for his genetics project in science. Kieran's data is shown.

The table shows the number of people who match each description.

Brown hair, Brown Eyes	Brown Hair, Blue Eyes	Blonde Hair, Brown Eyes	Blonde Hair, Blue Eyes
33	18	11	24

- a. Construct a two-way frequency table to summarize the data.

	<i>Brown Hair</i>	<i>Blonde Hair</i>	Total
<i>Brown Eyes</i>	33	11	44
<i>Blue Eyes</i>	18	24	42
Total	51	35	86

- b. The total number of students surveyed with brown eyes is 44.

This value represents a

- ☐ marginal frequency.
☐ joint frequency.

- c. The total number of students surveyed with brown eyes and blonde hair is 11.

This value represents a

- ☐ marginal frequency.
☐ joint frequency.



Monitor how students determine how to set up the columns and rows of the table.



How did you use the given joint frequencies to find the rest of the numbers in the table?



For students who need more of a challenge, consider having them create the joint frequency table using joint relative frequencies. They can then try to figure out how to complete the total column and row.

14.1.5: Try It: Constructing a Two-Way Table

Component Overview: Students work with their partner to complete two joint frequency tables.



14.1.5 Try It: Constructing a Two-Way Table

1. A survey of high school students asked participants their gender and if they play video games. The data is summarized:
 - 68 participants were males who play video games
 - 60 participants were males who do not play video games
 - 145 participants do not play video games
 - There were 279 participants in the survey

- a. Complete the joint frequency table.

Student Video Game Activity

	Play Video Games	Do Not Play Video Games	Total
Male	68	60	128
Female	66	85	151
Total	134	145	279

- b. How many female students do not play video games?
85 female students do not play video games.
- c. Of the students who play video games, how many were female?
Of the students who play video games, 66 are female.
- d. How many students surveyed play video games?
134 of the students surveyed play video games.



Monitor how students use the given data to complete the two-way frequency table. The way students find the missing numbers and place the given numbers can illuminate misconceptions students have about joint frequency tables.



What strategies or methods did you use in order to find all the numbers needed to complete the table?



For students who finish early, challenge them to summarize the data using different facts than those shared in question 1. Consider having them figure out if there is a way to summarize the data fully using less than four pieces of data from the table. Also, consider having students summarize the data using percentages, or relative frequencies, rather than integers.

14.1.5 Try It: Constructing a Two-Way Table (continued)

2. Jackson County Schools, located in Marianna, Florida, conducted a student survey to find out how students spend the majority of their free time outside of school.

The data of this survey is shown in tables below.

Elementary Students		
On the Phone	Playing Video Games	Reading
389	863	1,242

Middle School Students		
On the Phone	Playing Video Games	Reading
532	557	481

High School Students		
On the Phone	Playing Video Games	Reading
978	844	224

Construct a two-way frequency table that shows all of the survey data.

	On the Phone	Playing Video Games	Reading	Total
Elementary Students	389	863	1,242	2,494
Middle School Students	532	557	481	1,570
High School Students	978	844	224	2,046
Total	1,899	2,264	1,947	6,110



What do you notice about the table you made? How did you decide where to put the given numbers? How did you find the missing numbers?



Consider closing this component with a discussion about how two-way frequency tables are constructed or giving students the opportunity to check their work before moving on to 14.1.6 Wrap-Up section.



For a challenge, have students rewrite one or both of the two-way frequency tables using joint relative frequencies. They practiced this skill in 14.1.2 Guided Instruction, questions 11 and 12.

14.1.6 Wrap-Up: Reflection

Component Overview: Students complete a self-assessment based on their level of understanding of each key feature. It is designed to be done independently to gather specific, student-level data.



14.1.6 Wrap-Up: Reflection

1. Rate your confidence in creating two-way tables to display bivariate categorical data.

Answers will vary.



0 1 2 3 4 5 6 7 8 9 10

2. What are some questions you still have about two-way frequency tables?

Answers will vary.



Consider having students rate and keep track of their confidence ratings again near the middle and again at the end of Unit 14. Students can use these ratings to complete a reflection or self-assessment of their learning throughout Unit 14.